The Feller leg of the explosion dictionary: an escape diffusion with scale and speed read off the classical flow

Companion note to Theorem 6.1 (`thm:dictionary`)

The gap, and why H itself cannot carry the stochastic leg

Theorem 6.1(2)(c)–(d) of the paper asserts that when the classical time of flight

$$T(E) \;=\; \int^{\infty} \frac{\mathrm{d}x}{\sqrt{2\,(E-V(x))}}$$

is finite, "the regulator-independent object is the minimal (Feller) semigroup: a sub-stochastic, *explosive* process," and that one integral decides classical completeness, quantum essential self-adjointness, and "stochastic conservativity (Feller exit boundary)." The classical and quantum legs are rigorous; the stochastic leg is asserted without a construction. It cannot be carried by $H = \frac{1}{2}p^2 + V$ itself, for two independent reasons.

(a) Spectral. In the malicious case every self-adjoint extension of H has spectrum unbounded below (discrete, accumulating at $-\infty$), so e^{-tH} is not even a bounded semigroup, let alone sub-Markovian; a potential $V \to -\infty$ is neither a drift nor a nonnegative killing rate.

(b) Probabilistic. The standard bridge from a Schrödinger operator to a diffusion is the ground-state (Doob h-) transform $h^{-1}(H-E)h$, which requires a solution $h > 0$ of $(H-E)h = 0$ near $+\infty$. On the escaping end $E - V(x) \geq c > 0$, so $u'' = -2(E-V)u$ is oscillatory there (Sturm comparison: zeros spaced $\leq \pi/\sqrt{2c}$); *no* eventually-positive solution exists and no h-transform is available. The stochastic leg must therefore be carried by a different generator, and the classical mechanism dictates which one.

The escape diffusion

Fix E and x_0 with $\sup_{x \geq x_0} V(x) < E$ (possible since $V \to -\infty$), and on $I = [x_0, \infty)$ define the classical speed and clock

$$v(x) = \sqrt{2\,(E-V(x))} \;\geq\; \sqrt{2(E - \sup_I V)} > 0, \qquad \tau(x) = \int_{x_0}^{x} \frac{\mathrm{d}y}{v(y)}, \qquad T \equiv T(E) = \lim_{x\to\infty} \tau(x) \in (0, \infty].$$

$\tau : I \to [0, T)$ is a C^1 increasing bijection, and along the classical escape orbit $\dot{x} = v(x)$ one has $\dot{\tau} = 1$: τ is the coordinate in which the classical flow is translation at unit speed. The natural stochastic realization of the escape mechanism is the classical flow run with a Brownian-perturbed clock,

$$\mathrm{d}X_t \;=\; v(X_t) \circ \big(\mathrm{d}t + \varepsilon\, \mathrm{d}W_t\big) \qquad (\text{Stratonovich};\ \varepsilon > 0), \tag{1}$$

whose zero-noise skeleton is exactly $\dot{x} = v(x)$, together with its driftless variant $\mathrm{d}X_t = v(X_t) \circ \mathrm{d}W_t$ (Brownian motion in the classical clock). In the τ coordinate, (1) is literally Brownian

motion with unit drift and variance ε^2 on $[0,T)$, and the driftless variant is Brownian motion on $[0,T)$; explosion of X (arrival at $x=+\infty$) is the τ-process hitting the finite or infinite level T. In scale-and-speed form (which requires only V continuous, not C^1), the two variants are the regular diffusions on I with, respectively,

$$\text{drifted } (\beta=1): \quad s'(x)=\frac{e^{-2\tau(x)/\varepsilon^2}}{v(x)}, \quad m(\mathrm{d}x)=\frac{2}{\varepsilon^2}\,\frac{e^{2\tau(x)/\varepsilon^2}}{v(x)}\,\mathrm{d}x, \tag{2}$$

$$\text{driftless } (\beta=0,\ \varepsilon=1): \quad s(x)=\tau(x), \qquad m(\mathrm{d}x)=\frac{2\,\mathrm{d}x}{v(x)} \;=\; 2\big[2(E-V(x))\big]^{-1/2}\mathrm{d}x, \tag{3}$$

with killing measure zero, reflection at the (regular) left endpoint x_0, and the minimal (absorbing) convention at $+\infty$. One checks $D_m D_s=\frac{\varepsilon^2}{2}(v\,\partial_x)^2+\beta\,v\,\partial_x$, i.e. the generator of (1).

Two identifications make this the *canonical* choice rather than a convenient one. First, the driftless speed measure (3) is exactly (twice) the WKB deficiency density of the paper's one-line identity (its Eq. `eq:onelineidentity`, $|\psi_\pm|^2\approx[2(E-V)]^{-1/2}$): $m(\mathrm{d}x)=2\,|\psi_\pm^{\mathrm{WKB}}(x)|^2\,\mathrm{d}x$, so the speed-measure mass near $+\infty$ *is* the deficiency-solution norm, $m((x_1,\infty))=2\int_{x_1}^{\infty}\mathrm{d}x/v\approx 2\int_{x_1}^{\infty}|\psi_\pm|^2\,\mathrm{d}x$. Second, the scale mass near $+\infty$ in (3) is $T(E)$ itself. Both Feller data are the one classical integral.

The lemma

Boundary-classification notation (Feller [1, 2]; Itô–McKean [3]; Karlin–Taylor [4], Ch. 15.6; Borodin–Salminen [5], Ch. II.1): with $S(x_0,\xi]=s(\xi)-s(x_0)$, $M(x_0,\xi]=m((x_0,\xi])$,

$$\Sigma=\int_{x_0}^{\infty}M(x_0,\xi]\,\mathrm{d}S(\xi), \qquad N=\int_{x_0}^{\infty}S(x_0,\xi]\,\mathrm{d}M(\xi);$$

$+\infty$ is *regular* iff $\Sigma<\infty$ and $N<\infty$ (equivalently S,M both finite), *exit* iff $\Sigma<\infty$, $N=\infty$, *entrance* iff $\Sigma=\infty$, $N<\infty$, *natural* iff both diverge; it is *accessible* (attainable in finite time with positive probability) iff $\Sigma<\infty$, i.e. iff regular or exit.

Lemma 1 (Feller leg of the explosion dictionary)**.** *Let V satisfy the hypotheses of Theorem 6.1 (C^1, $V\to-\infty$ eventually monotonically, $|V'||V|^{-3/2}$ bounded); in fact only V continuous with $V(x)\to-\infty$ is used below. Fix E, x_0, v, τ, $T=T(E)$ as above, and let X be either escape diffusion* (2) *($\beta=1$, any $\varepsilon>0$) or* (3) *($\beta=0$), reflected at x_0, minimal at $+\infty$; let $\zeta=\inf\{t: X_{t-}=+\infty\}$ be its explosion time and $(P_t^{\min})$ the associated minimal semigroup on $C_0(I)/L^\infty(I)$. Then:*

(i) ***(classification)*** *$+\infty$ is a* regular *boundary if $T(E)<\infty$ and a* natural *boundary if $T(E)=\infty$; it is never exit and never entrance. Explicitly, for $\beta=0$ the two Feller test integrals coincide and equal the square of the classical time of flight,*

$$\Sigma \;=\; N \;=\; T(E)^2,$$

while for $\beta=1$, $\Sigma=T-\frac{\varepsilon^2}{2}\big(1-e^{-2T/\varepsilon^2}\big)$ and $N=\frac{\varepsilon^2}{2}\big(e^{2T/\varepsilon^2}-1\big)-T$, both finite iff $T<\infty$. The classification is independent of E, of $\beta\in\{0,1\}$, and of $\varepsilon>0$.

(ii) ***(explosion $\Leftrightarrow$ $T < \infty$)*** *If $T(E) < \infty$ then $\mathbb{P}_x(\zeta < \infty) = 1$ for every $x \in I$: the minimal semigroup is strictly sub-stochastic ($P_t^{\min}\mathbf{1} < 1$ for all $t > 0$), and probability reaches $+\infty$ in finite time. The mean explosion time from x_0 is the accessibility integral itself,*

$$\mathbb{E}_{x_0}[\zeta] \;=\; \Sigma \;=\; \begin{cases} T(E)^2, & \beta = 0, \\ T(E) - \frac{\varepsilon^2}{2}\big(1 - e^{-2T(E)/\varepsilon^2}\big) \;\in\; \big(T - \frac{\varepsilon^2}{2},\, T\big], & \beta = 1, \end{cases}$$

and in the small-noise limit of the drifted model, $\zeta \to T(E)$ almost surely as $\varepsilon \downarrow 0$: the minimal process loses its mass at the classical escape time. If $T(E) = \infty$ then $\mathbb{P}_x(\zeta < \infty) = 0$: the semigroup is conservative ($P_t^{\min}\mathbf{1} = \mathbf{1}$) and, $+\infty$ being natural, it is the unique *Feller semigroup generated by $\frac{\varepsilon^2}{2}(v\partial_x)^2 + \beta v\partial_x$ with reflection at x_0: no lateral condition can be imposed at infinity [1, 2].*

(iii) ***(dictionary)*** *Consequently one integral decides all three legs: $T(E) < \infty \Leftrightarrow$ classical escape in finite time $\Leftrightarrow$ (under the theorem's regularity, by its legs (1)–(2)) limit circle at $+\infty \Leftrightarrow$ $+\infty$ regular and X explosive; $T(E) = \infty \Leftrightarrow$ limit point $\Leftrightarrow$ $+\infty$ natural and X conservative and unique. Moreover the correspondence is structural, not merely numerical: for $\beta = 0$ the diffusion's own Weyl classification agrees ($u = 1$ and $u = s$ solve $D_m D_s u = 0$, and $\int \mathrm{d}m = 2T$, $\int s^2 \mathrm{d}m = \frac{2}{3}T^3$, so $D_m D_s$ is limit circle at $+\infty$ iff $T < \infty$), the speed measure is the WKB deficiency density (the paper's Eq.* `eq:onelineidentity`*), and at a regular boundary Feller's lateral conditions at $+\infty$ (absorbing, reflecting, elastic, sticky, jump-in [2, 3]) parametrize the sub-Markov extensions of the minimal semigroup—the stochastic mirror of the quantum $U(1)$ family of Theorem 6.1(2)(a), with the conservative lateral conditions mirroring the completions that return probability from infinity (Theorem 6.1(2)(b)).*

(iv) ***(exit is impossible)*** *For* any *regular diffusion on (x_1, ∞) whose speed measure has finite mass near $+\infty$—in particular any diffusion honoring the WKB identity $m(\mathrm{d}x) \propto |\psi_\pm|^2 \mathrm{d}x$ with $T < \infty$—the boundary $+\infty$ cannot be exit in Feller's strict sense: accessibility forces $S < \infty$, and then $N \le S \cdot M < \infty$, so an accessible boundary with finite speed mass is regular. "Exit" in Theorem 6.1(2)(d) must therefore be read as* accessible *(regular), not as Feller's exit class.*

Proof. Throughout, transport everything to the clock coordinate: the map $\tau : I \to [0, T)$ is a C^1 increasing homeomorphism (v is continuous and bounded below by $v_{\min} = \sqrt{2(E - \sup_I V)} > 0$), scale functions and speed measures are intrinsic under such maps (Itô–McKean [3], §4.1–4.6; Rogers–Williams [6], V.46–47), and the image data of (2)–(3) under τ are exactly the scale and speed of Brownian motion on $[0, T)$ with drift β and variance ε^2: $\mathrm{d}\tilde{s}(u) = e^{-\varrho u}\mathrm{d}u$, $\tilde{m}(\mathrm{d}u) = \frac{2}{\varepsilon^2}e^{\varrho u}\mathrm{d}u$ with $\varrho = 2\beta/\varepsilon^2$ (substitute $u = \tau(x)$, $\mathrm{d}x = v\,\mathrm{d}u$). Hence $X_t = \tau^{-1}(Z_t)$ in law, where Z is that linear diffusion on $[0, T)$, reflected at 0, absorbed at T, with the same time parameter and $\zeta = \inf\{t : Z_t = T\}$. All four classification integrals may therefore be computed for Z; the Lebesgue–Stieltjes substitution $u = \tau(\xi)$ shows they equal the corresponding integrals in x.

(i). Case $T < \infty$: on the bounded interval $(0, T)$ both $\tilde{S}(0, T) = \int_0^T e^{-\varrho u}\mathrm{d}u$ and $\tilde{M}(0, T) = \frac{2}{\varepsilon^2}\int_0^T e^{\varrho u}\mathrm{d}u$ are finite, hence $\Sigma \le \tilde{S}\,\tilde{M} < \infty$ and $N \le \tilde{S}\,\tilde{M} < \infty$: $+\infty$ is regular. For $\beta = 0$, $\varepsilon = 1$: $\tilde{S}(0, \xi] = \xi$, $\tilde{M}(0, \xi] = 2\xi$, so $\Sigma = \int_0^T 2\xi\,\mathrm{d}\xi = T^2 = N$. For $\beta = 1$: $\tilde{M}(0, \xi] = e^{\varrho\xi} - 1$ ($\varrho = 2/\varepsilon^2$; the prefactors cancel, $\frac{2}{\varepsilon^2} \cdot \frac{1}{\varrho} = 1$), $\mathrm{d}\tilde{S} = e^{-\varrho\xi}\mathrm{d}\xi$, whence $\Sigma = \int_0^T (1 - e^{-\varrho\xi})\mathrm{d}\xi = T - \frac{\varepsilon^2}{2}(1 - e^{-2T/\varepsilon^2})$

and $N = \int_0^T \frac{1-e^{-\varrho\xi}}{\varrho} \cdot \frac{2}{\varepsilon^2} e^{\varrho\xi} \mathrm{d}\xi = \int_0^T (e^{\varrho\xi} - 1)\mathrm{d}\xi = \frac{\varepsilon^2}{2}(e^{2T/\varepsilon^2} - 1) - T$. Case $T = \infty$: for $\beta = 0$, $\Sigma = N = \lim_{T' \to \infty} T'^2 = \infty$; for $\beta = 1$, $\Sigma = \int_0^\infty (1 - e^{-\varrho\xi})\mathrm{d}\xi = \infty$ and $N = \int_0^\infty (e^{\varrho\xi} - 1)\mathrm{d}\xi = \infty$: natural. Since $\Sigma < \infty \Leftrightarrow T < \infty \Leftrightarrow N < \infty$ in all cases, the exit and entrance classes never occur. E-independence: if $V \to -\infty$ then for $E \neq E'$, $(E - V(x))/(E' - V(x)) \to 1$ as $x \to \infty$, so $T(E) < \infty \Leftrightarrow T(E') < \infty$ and (i) applied at E' gives the same class.

(ii). Let $T < \infty$. For $\beta = 1$ the Skorokhod decomposition of the reflected process gives $Z_t = z + t + \varepsilon W_t + L_t \geq z + t + \varepsilon W_t \to \infty$ a.s., so Z hits T a.s.; for $\beta = 0$, Z is reflecting Brownian motion, which is recurrent on $[0, \infty)$ and hits T a.s. Hence $\mathbb{P}_x(\zeta < \infty) = 1$, and since Z has everywhere-positive transition density on $(0, T)$ before absorption and T is accessible, $\mathbb{P}_x(\zeta \leq t) > 0$ for every $t > 0$, i.e. $P_t^{\min}\mathbf{1}(x) = \mathbb{P}_x(\zeta > t) < 1$ for all $t > 0$. For the mean, solve $D_{\tilde{m}} D_{\tilde{s}} u = -1$ on $(0, T)$ with $D_{\tilde{s}} u(0) = 0$ (reflection) and $u(T) = 0$: $D_{\tilde{s}} u(\xi) = -\tilde{M}(0, \xi]$, so $u(z) = \int_z^T \tilde{M}(0, \xi]\, \mathrm{d}\tilde{S}(\xi)$ (Karlin–Taylor [4], §15.3), and at $z = 0$ this is exactly the integral Σ evaluated in (i); Dynkin's formula applies since ζ has finite mean by the explicit bound. This yields $\mathbb{E}_{x_0}[\zeta] = T^2$ ($\beta = 0$) and $\mathbb{E}_{x_0}[\zeta] = T - \frac{\varepsilon^2}{2}(1 - e^{-2T/\varepsilon^2})$ ($\beta = 1$). Small noise: with $Z_0 = 0$, Skorokhod gives the sandwich $t + \varepsilon W_t \leq Z_t \leq t + \varepsilon W_t + \sup_{r \leq t}\big(-(r + \varepsilon W_r)\big)^+$, and $\sup_{t \leq T+1} \varepsilon |W_t| \to 0$ a.s., so $\zeta \to T$ a.s.

Let $T = \infty$. By (i), $+\infty$ is natural, hence inaccessible: $\Sigma = \infty$ implies $\mathbb{P}_x(\zeta < \infty) = 0$ (Feller's test for explosions; [4], Lemma 15.6.2, or [3], §4.2). With reflection at the regular endpoint x_0 and no killing, the lifetime is a.s. infinite and $P_t^{\min}\mathbf{1} = \mathbf{1}$. Uniqueness at a natural boundary is Feller's theorem: the generator's domain requires no lateral condition at a natural endpoint and admits none, so the sub-Markov semigroup with the given (reflecting) condition at x_0 is unique [1, 2].

(iii). The chain of equivalences is the conjunction of (i), (ii) with the classical statement ($\dot{x} = v$ reaches $+\infty$ at time exactly $T(E)$, the $\varepsilon \to 0$ limit above) and with legs (1)–(2) of Theorem 6.1 (Weyl alternative and limit-point/limit-circle criteria, Reed–Simon [7], Thms. X.7–X.10, under the theorem's regularity hypothesis). The Weyl computation for $D_m D_s$ at $\beta = 0$ is elementary: the solutions of $D_m D_s u = 0$ are $u = 1$ and $u = s = \tau$, and near $+\infty$ $\int \mathbf{1}^2\, \mathrm{d}m = 2T$, $\int s^2\, \mathrm{d}m = 2\int_0^T u^2 \mathrm{d}u = \frac{2}{3}T^3$, both finite iff $T < \infty$ (limit-point/limit-circle for $D_m D_s$ is λ-independent as usual). The identification $m(\mathrm{d}x) = 2|\psi_\pm^{\mathrm{WKB}}|^2 \mathrm{d}x$ is Eq. (3) against the paper's Eq. `eq:onelineidentity`. The lateral-condition statement at a regular boundary is Feller's classification of boundary conditions [2] (see also [3], §3.7–3.8, [5], II.1.7).

(iv). Suppose $+\infty$ is accessible, i.e. $\Sigma < \infty$. If $S(x_1, \infty) = \infty$ then for any interior ξ_0, $\Sigma \geq M(x_1, \xi_0] \int_{\xi_0}^\infty \mathrm{d}S = \infty$, a contradiction; so $S < \infty$. Then $N = \int S(x_1, \xi]\, \mathrm{d}M(\xi) \leq S(x_1, \infty)\, M(x_1, \infty)$, which is finite whenever the speed mass $M(x_1, \infty)$ is finite. Hence $N < \infty$, and the boundary is regular, not exit. □

Remark 1 (hypotheses audit). The lemma uses strictly less regularity than Theorem 6.1: only V continuous and eventually $< E$ (plus $V \to -\infty$ for the E-independence claim). The eventual monotonicity and the bound $|V'||V|^{-3/2}$ are needed only for the *quantum* leg (the limit-point/limit-circle criteria cited in the theorem); the Stratonovich form (1) uses $V \in C^1$, which the theorem already assumes, and the scale–speed construction (2)–(3) does not need even that. No hypothesis had to be added.

Remark 2 (regular, not exit, is the physically correct class). The strict Feller exit class would *forbid* re-entry from the boundary, contradicting the paper's own Theorem 6.1(2)(b) and the revival numerics of its explosion figure (`fig:explosion`; completions returning $\approx 95\%$ of the probability from infinity). The regular class is exactly right: the minimal (absorbing) lateral

condition gives the sub-stochastic explosive semigroup of (2)(c)—regulator-independent, as in the left panel of the figure—while the conservative lateral conditions (reflecting, sticky, jump-in) are the stochastic counterparts of the $U(1)$ family of self-adjoint extensions in (2)(a)–(b), including a jump-in condition $+\infty \to$ interior mirroring the "wormhole" completion. So the correction strengthens the dictionary rather than weakening it.

Remark 3 (proposed rewording in the paper). Minimal edits that make leg (3) a theorem: in (2)(c), after "the minimal (Feller) semigroup [1]", specify the process, e.g.: "*of the escape diffusion* $\mathrm{d}X = \sqrt{2(E - V(X))} \circ (\mathrm{d}t + \varepsilon\, \mathrm{d}W)$*—the classical flow run with a Brownian-perturbed clock, whose speed measure is the WKB density of Eq.* `eq:onelineidentity`—a sub-stochastic, explosive process losing mass at the classical escape time (mean explosion time $= T(E) + O(\varepsilon^2)$, and $\to T(E)$ a.s. as $\varepsilon \to 0$)"; and in (2)(d), replace "(Feller exit boundary)" by "(Feller boundary classification: $+\infty$ is regular—accessible—iff $T < \infty$, natural iff $T = \infty$)". The abstract's item (vii) needs no change beyond reading "stochastic conservativity" through this lemma.

VERDICT: PROVABLE-WITH-RESTATEMENT. Leg (3) of Theorem 6.1 is rigorously provable once (a) the stochastic object is named—the escape diffusion (1)/(2)–(3), not any semigroup of H—and (b) "Feller exit boundary" is corrected to "accessible (regular) boundary": for this diffusion, $+\infty$ is regular iff $T(E) < \infty$ (minimal semigroup sub-stochastic and explosive, mean lifetime the classical time of flight up to $O(\varepsilon^2)$) and natural iff $T(E) = \infty$ (conservative and unique). Strict "exit" is provably impossible for any diffusion consistent with the paper's own WKB identity, and would contradict the paper's return-from-infinity completions; "regular" is both correct and a better fit to the theorem's (2)(a)–(b).

References

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